Turning Sugarcane Waste into Value

A Grounded Survey of Hawaiian Practice, Microbial Conversion Routes, and Revenue Options for Kenya's Bagasse Surplus

Prepared for a Kenyan sugar-cane grower and fibre supplier

October 2026

*Every figure, price and yield in this document is attributed to a specific retrieved source, numbered in square brackets and listed in full at the end with the exact sentence it rests on. Where a claim could not be grounded in a retrieved source, it is marked **[unverified]**. Nothing here is a recommendation to spend money: it is a survey of what has been demonstrated, what has failed, and where the money actually is.*

1 The situation in one page

A sugar factory takes in cane and separates it into juice and fibre. The juice becomes sugar; the fibre — bagasse — is what we are talking about here. It is not a small residue: bagasse *"constitutes up to 28% of the weight of crushed sugarcane"*[8], and a parallel engineering description puts it as *"about 30% of the weight of cane"*[86]. For a mill crushing 3,000 tonnes of cane a day, that is on the order of 900 tonnes of wet fibre every single day of the crushing season.

The instinct that "bio transformation is a good way to get value out of waste" is correct, and it is also precisely how Hawaii's industry discovered where the limits are. Hawaii ran commercial sugar for more than 150 years and closed its last plantation on Maui at the end of 2016 — and the reason it survived as long as it did, past the point where sugar alone made money, was the fibre.

> *"HC&S generates electricity by burning the fiber known as bagasse left over from processed cane, as well as coal, in a boiler to power its sugar mill and irrigation pumps."*[100]

> *"As we've been saying – electricity sales is why A&B kept farming sugar long after it became difficult to make a profit."*[100]

That is the central lesson of the Hawaiian experience, and it is uncomfortable: the byproduct was not a bonus, it was the reason the business existed at all in its final years. When the power purchase agreement was amended, the owner decided *"to throw in the towel and close the 36,000-acre Hawaiian Commercial & Sugar Co. farm on Maui by year's end"*[100], laying off 675 people[100]. A 36,000-acre operation — with its own boilers, its own power contract and 145 years of practice — could not make the arithmetic work once the electricity revenue was withdrawn.

The second lesson is that the amount of material is not the constraint. In 2006 Hawaii's sugar industry generated, as measured by the state's own biofuel assessment:

> *"In 2006, HC&S produced about 430,000 tons of bagasse at 50 percent moisture content (215,000 tons of fiber) and 60,000 tons of molasses"*[117]

with cane trash adding a further 110,000 tons of fibre, and the smaller Gay & Robinson mill on Kauaì contributing another 100,000 tons of bagasse and 30,000 tons of cane trash.[117] The state assessment named the whole family of feedstocks explicitly: *"municipal solid waste, sewage sludge, FOG, bagasse fiber, cane trash, molasses, pineapple processing waste, macadamia nut shells,*

animal manures, and landfill gas"[117]. It also ranked them: *"The sugar industry is the largest single source of agricultural biomass residues in the State of Hawai'i."*[117]

So the question your friend is really asking is not "what can we do with bagasse" — the answer to that is a long list, and it is below — but "which of these things makes money at our scale, with our capital, and with our market within trucking distance." Hawaii's answer, in the end, was: electricity first, and everything else struggled.

2 What Hawaii actually did

2.1 Bagasse electricity, and why it was also the weak point

Hawaii's mills burned bagasse in boilers to raise steam, driving both the mill and a generator, then sold surplus power to the utility. HC&S sold to Maui Electric, which *"once relied on HC&S for about 10 percent of its electricity supply"*[100]. The volumes were real and recorded: *"HC&S provided 1,265.5 megawatt hours of power to Maui Electric Co. in the third quarter"*[104], having previously delivered *"as much as 12 MW of power year-round"*[104] before the agreement was scaled back — a contraction that directly preceded the closure.

The structural problem is visible in the same reporting: HC&S burned coal alongside bagasse, *"In 2014 the company burned 57,100 tons of coal"*[100], because bagasse alone could not carry the load year-round — a mill only has fresh bagasse during the crushing season. Kenya's mills hit exactly the same wall, and we have a documented Kenyan case of it.

2.2 Molasses, rum, and the value of a finished bottle

Hawaii's molasses stream was large (60,000 tons from HC&S in 2006 alone)[117] and historically fed both cattle and, later, a spirits industry. Kōloa Rum Company on Kauaì has distilled since its first batch in 2009[139] and sells the finished bottle at a multiple of any bulk molasses price — which is the general point about every "upgrade" route in this document: value multiplies with each processing step you take ownership of.

2.3 Ethanol from fibre: attempted, and instructive

Hawaii tried to convert fibre and cane trash into ethanol. The state's biofuel assessment assessed *"bagasse fiber, cane trash, molasses"*[117] as feedstocks and the island of Kauaì became the test case: Gay & Robinson pursued ethanol from its fibrous byproducts rather than from sugar or molasses[115], and ultimately the last cane was harvested in 2010 with the proposed ethanol plant handed to another developer[132]. The technology was real; the economics were not, at that scale, at that time.

2.4 What replaced sugar on the land

After closure, HC&S's parent moved the land to *"ranching and diversified agriculture"*[154]. Cane land did not become an industrial feedstock farm; it became pasture and diversified crops. For your friend, the implication is that the fibre is best treated as a *byproduct of an ongoing process*, not as the product of a dedicated operation.

2.5 The Hawaii summary table

Route	Hawaii outcome	Evidence
Bagasse cogeneration sold to grid	Central to survival; collapsed when the PPA was amended	[100][104]
Molasses to rum	One commercial distillery, still operating	[139]
Fibre/trash to ethanol	Attempted at Gay & Robinson; plant did not proceed	[115][132]
Coal co-firing in mill boilers	57,100 tons of coal burned in 2014 — bagasse alone was insufficient	[100]
Land after sugar	Pasture and diversified agriculture	[154]

3 The Kenyan landscape

The good news is that Kenya's own numbers are already documented, and the government's own policy documents describe cogeneration as a national opportunity.

> *"an estimated 38 MW capacity is installed in Kenya's six operational sugar factories which use bagasse to generate electricity"*[49]

The flagship case is Mumias. *"Mumias Sugar Company Ltd (MSC) generates electricity from its bagasse-based co-generation plant"*[51], and at *"about 34 MW of electricity, 26 MW of which is sold to Kenya Power and Lighting Company (KPLC) based on a long-term power purchase agreement (PPA), while the remaining is used for factory"* operations[51] — with *"100 tons/day of ash"*[51] as a secondary output. That is the single most important precedent available to your friend: a Kenyan mill sold 26 MW into the national grid on a PPA.

Then the cautionary half of the same story, from the peer-reviewed analysis of that plant: *"It was established that the sugar factory invested in a 34 MW cogeneration power plant to export 26"* MW[56], but *"Reliance on bagasse alone as the fuel placed the plant at the mercy of the"* crushing season, and the study records that the *"cogeneration power plant became unsustainable and export to the grid was stopped"*[56]. The lesson is specific and worth repeating: **a bagasse-only power plant is seasonal and fragile**. It needs a second fuel (coal, wood waste, or a stored, densified form of bagasse) to run outside the crush.

The economics at smaller scale are documented too. A study of an operating traditional factory found *"the 3,000 TCD factory can invest in a 15 MW power plant based on current average factory performance indicators"*[57] — useful because 3,000 tonnes of cane per day is a realistic Kenyan mill size, and it suggests the realistic export is a fraction of Mumias's, not zero.

What Kenya currently leaves on the table is stated bluntly by the industry's own association. Diversification is available — *"ethanol, cogeneration (electricity from bagasse), briquettes (charcoal from bagasse), paper and packaging, filter mud (press mud), and organic fertilizer from boiler ash"*[53] — but *"most Kenyan sugar mills remain unprepared to capitalize on these opportunities"*[53]. Two further facts matter for demand: the Sugar Act 2024 is expected to compensate farmers for by-products such as bagasse and molasses *"from which they have previously not benefitted"*[53], and new capacity is being built with byproducts designed in from the start — one new estate project *"targets 120,000 tonnes of sugar a year, about 13 million litres of ethanol and 25 megawatts of*

electricity from bagasse"[54].

And the waste problem is real, not theoretical. Excess bagasse is *"dumped on land around sugar mills and left to rot, releasing methane and polluting local waterways"*[125].

4 The option list, route by route

Each route below gives: what it is, what the evidence says it produces, what it needs, and an honest read on whether it fits a Kenyan fibre supplier.

4.1 Route 1 — Densify it and sell it as fuel

What: dry and compress bagasse into briquettes or pellets, optionally carbonised into charcoal-substitute.

Evidence: A Kenya-facing research partnership has already produced hardware for exactly this pre-step: the machine is *"capable of drying sugarcane waste, known as bagasse, at a rate of two tonnes an hour"*[125], described as *"a working prototype"*[125] for converting bagasse into clean cooking fuel. Commercial carbonised-bagasse charcoal has been made in Kenya before: a plant at the Chemelil Sugar Factory was commissioned with sales under the brand "CaneCoal"[127]. The FAO also documents the upstream case for carbonised bio-briquettes from bagasse as a substitute for forest charcoal in household cooking[124].

Needs: drying (the biggest cost, and Kenya has sun), a press or briquetting mill, and a distribution channel into the same market that sells charcoal — cookstove fuel is a mature, cash market in Kenya.

Read: *This is the best fit for a fibre supplier rather than a mill.* It needs no fermentation, no sterile process, no chemical plant; it turns a wet liability into a dry, storable, transportable commodity that solves the seasonality problem that killed Mumias's grid export. It is also the lowest-technology route with a proven Kenyan precedent.

4.2 Route 2 — Black soldier fly larvae: protein from fibre

What: grow *Hermetia illucens* larvae on organic waste; harvest the larvae as animal feed protein and the residue (frass) as fertiliser.

Evidence: BSF larvae *"can convert biowaste and by-products into body mass high in protein (∼40% dry matter, DM) and lipid (∼30% DM)"*[3]. Protein content in practice ranges from *"30.6% to 52.2%"*[4]. On sugarcane byproducts specifically, a direct comparison of substrates found that although canteen leftovers performed best, *"the combination of bagasse and vinasse (BV) proved to be the most suitable substrate among the sugarcane by-products for BSF larva development"*[2] — i.e. bagasse works, and bagasse *plus vinasse* works better than bagasse alone.

The Kenyan demand case is the strongest part of this route:

> *"For many years, the cost of commercial feeds for poultry and pig farming has been increasing, discouraging farmers from investing in these ventures."*[89]

> *"Currently Kenya does not produce enough quantities of soybean, and limited local supply necessitates importation, often making the commodity unaffordable, especially for small-scale farmers."*[89]

The frass byproduct is itself a saleable fertiliser, with Kenyatta University research reporting that *"The BSF frass fertilizer can perform equally good as commercial fertilizers in terms of maize yield and nitrogen use efficiency"*[91]. BSF is already an established business in Kenya — ICIPE began training SMEs in rearing from 2015[92] — and awareness among farmers is still low, which is both an obstacle and an opportunity[90].

Needs: a rearing facility (shade, humidity control, clean water), consistent organic feedstock, and — the real constraint — a buyer for dried larvae meal. Bagasse alone is too lignified for good larval growth; it must be blended with a nitrogen-rich stream (vinasse, kitchen waste, manure).

Read: *The best fit if your friend is near poultry farmers and can secure a second feedstock.* It matches the existing local protein deficit, has a Kenyan research and training base, and produces two saleable products (larvae meal and frass). It is also the route where bagasse is a *co-substrate* rather than the whole feedstock — which is exactly what the evidence says works.

4.3 Route 3 — Ruminant feed (with treatment)

What: feed bagasse to cattle, either untreated (mostly filler) or after alkali/urea treatment to raise digestibility.

Evidence: The FAO's feed volume is unambiguous that raw bagasse is poor: *"its low digestibility and nitrogen content limited its use to filler in concentrates or at very low inclusion levels"*[43]. The established fix is alkali or urea treatment, which raises digestibility toward that of better roughages[43]. The FAO also documents a proven Kenyan-adjacent practice of feeding whole sugarcane to crossbred cattle producing 10–12 litres of milk[48], and urea treatment has been studied specifically on bagasse in Latin-square cattle trials[45].

Needs: urea or alkali, water, airtight storage for ensiling, and cattle within reach. The treatment step is cheap but must be done correctly — urea-treated feed is toxic if badly managed.

Read: *A route for a friend who already keeps cattle or has a dairy cooperative nearby.* It is low-capital and improves drought resilience, but bagasse is a maintenance feed, not a growth feed.

4.4 Route 4 — Anaerobic digestion: biogas, methane, hydrogen

What: digest bagasse (often blended with mill wastewater, molasses, or press mud) in an anaerobic reactor to produce methane-rich biogas for heat, power, or cooking.

Evidence: Sugarcane processing wastes are a recognised AD feedstock across first- and second-generation streams[9], and AD of pretreated industrial residues is well established[10]. Bagasse's lignin is the obstacle, which is why study focuses on pretreatment[11][12]. One Griffith University study is directly relevant to feasibility, examining methane production from bagasse under wet, hemi-solid, and solid-state conditions[8] — solid-state digestion matters in Kenya because it needs far less water.

Needs: a sealed reactor, water balance, temperature control, and — for anything beyond farm-scale cooking gas — gas handling and a generator.

Read: *Likely the weakest fit as a standalone business at small scale*, because the lignin in bagasse resists digestion without pretreatment, and pretreatment costs money. It becomes attractive as a *companion* to a mill that already has wastewater and press mud to blend in.

4.5 Route 5 — Biochar: carbon, soil, filtration

What: pyrolyse bagasse with limited oxygen into a carbon-rich solid; sell as soil amendment, as a filtration/adsorbent material, or as a fuel.

Evidence: Bagasse biochar is well reviewed[18], and field evidence exists that it raises yield and soil carbon in cultivated systems[14]. The temperature of pyrolysis determines the product's properties[18]. As an adsorbent — for heavy metals and dyes in water — bagasse-derived carbon is repeatedly demonstrated[65][66], with one study reporting *"a maximum adsorption efficiency of 95%"* for methylene blue using bagasse pulp[68].

Needs: a kiln or retort (retort preferred — it captures rather than wastes the volatiles), feedstock drying, and either a farm-input market or a water-treatment buyer.

Read: *Strong second route alongside briquettes*, because the same kiln can produce both. Soil amendment sells locally to farms; the water-treatment market is real but demands consistent quality and is harder to enter from a small operation. This route also has a carbon-credit story — mark it **[unverified]** until a specific verified methodology and buyer is identified, because carbon market access is where claims outrun contracts.

4.6 Route 6 — Enzymatic / second-generation bioethanol

What: hydrolyse bagasse's cellulose to sugars, ferment to ethanol.

Evidence: The yields are documented and substantial. A whole-cane 2G study found that with maximum diversion of bagasse to ethanol, *"an increase of 25.8% in ethanol production (reaching 115.2 L/tonne of sugarcane) was achieved"*[37]. A review of the field notes that with pretreatment, conversion can reach up to *"90% conversion of sugars to ethanol compared to 20% without pretreatment"*[41] — the entire economics of this route live in the pretreatment step. Hydrothermal and ensiling-based pretreatments have been optimised empirically[40], and yeasts isolated for the purpose can ferment the resulting hydrolysates[38].

Needs: pretreatment plant (usually steam-explosion or dilute acid), enzyme supply (a major recurring cost), fermentation vessels, and distillation. This is an industrial chemical plant, not a farm operation.

Read: *Not the right first move for a fibre supplier.* The yields are real; so are the capital costs and the enzyme bill. Note that Kenya is *already* building this at estate scale — the new 50,000-acre project targeting *"13 million litres of ethanol and 25 megawatts of electricity"*[54]. Competing head-on with a state-backed estate is a poor position; supplying fibre to one may not be.

4.7 Route 7 — Sugarcane wax, policosanol and octacosanol

What: extract the wax from press mud / filter mud and sell high-value long-chain alcohols.

Evidence: Sugarcane wax is extracted from press mud, a mill waste[71], and the long-chain fatty alcohol octacosanol is its main constituent, extractable from filter mud containing *"6.85 g/100 g waxes"*[75]. Reported recovery of the most abundant aliphatic alcohol via one method is *"17.34%"*[70], and the compounds are attributed cholesterol-lowering and related biological functions[70][71].

Needs: solvent extraction (ethanol or supercritical CO_2), press mud access — which is the mill's, not the fibre supplier's — and a nutraceutical buyer. Regulatory status of policosanol health claims

differs by market.

Read: *High value, wrong feedstock owner.* This is a mill-side opportunity, and it is the one route where the material is press mud rather than bagasse. Worth raising with the mill, not pursuing alone.

4.8 Route 8 — Materials: pulp, paper, board, composites, nanocellulose

What: use the fibre as-is for board and composites, or refine it to pulp, paper, or cellulose nanocrystals.

Evidence: Bagasse has long been used for pulp and paper and for composition panels — FAO's Forest Products Laboratory line of work treats bagasse as one of the non-wood fibres used worldwide for these purposes[31][34][35], with practical composite-board work done on bagasse-reinforced panels[33] and particleboard reviewed recently[36]. At the high end, cellulose nanocrystals are being produced from bagasse through optimised acid hydrolysis[78], with the overall field — extraction, films, packaging — reviewed and expanding[76][77].

Needs: For board: a press, resin, drying, and a builder/furniture market. For pulp: a chemical pulping line (large capital). For nanocellulose: laboratory-grade chemistry and equipment.

Read: *Board and composites are plausible locally; paper and nanocellulose are not, at supplier scale.* Boards suit a small operation serving a local construction or furniture trade. Note that the FPL work also cautions that *"composition panels made from agricultural fibers are of somewhat poorer quality than those made from wood"*[34] unless blended — which is a real product-development constraint, not a detail.

4.9 Route 9 — Mushrooms on bagasse

What: grow oyster mushrooms (*Pleurotus*) on pasteurised bagasse as substrate.

Evidence: Bagasse is repeatedly identified as a suitable substrate, with one trial reporting *"the highest yield (120.00g)... obtained from the treatment of 80% of sugarcane bagasse supplemented with 20% wheat bran"*[21] and fast mycelial colonisation[20]. Trials comparing substrates show performance varies by species and supplement, with paddy straw sometimes ahead[19][22].

Needs: pasteurisation (steam or hot water), spawn, and — crucially — a humid, dark fruiting room and a fresh-produce market. Spent substrate afterwards is still a feed/compost input.

Read: *A good smallholder route with quick cash cycles*, but it is a food business (perishable, market-dependent) rather than a waste-processing business. Worth piloting at household scale before scaling.

4.10 Route 10 — Vermicompost

What: feed pre-rotted bagasse to earthworms; sell the castings.

Evidence: Vermicomposting of bagasse with *Eudrilus eugeniae* is documented[151], with one study reporting that *"Degradation of hard fibrous sugarcane bagasse resulted relatively in a higher vermicompost production (55.79 and 70.34%)"*[149], and combination with biochar has been studied for combined biowaste management[153].

Needs: shade, moisture, worm stock, several weeks of patience, and a gardener/farmer market.

Read: *Nice complementary route, low capital, low margin.* Best paired with BSF (the frass and the castings sell into the same farm-input channel).

4.11 Route 11 — Adsorbents and activated carbon

Already covered in the biochar section as a product variant; noted separately here because the buyer is different. Bagasse-derived activated carbon via chemical activation[65] and bagasse biofilters for metal and dye removal[66][69] are demonstrated, with high removal efficiencies reported[68]. **Read:** a value-added outlet for a biochar operation if a water-treatment or textile-effluent buyer exists locally; otherwise the soil market is easier.

4.12 Route 12 — Poultry litter / bedding

What: use dried chopped bagasse as poultry bedding.

Evidence: Sugarcane bagasse has been evaluated as a litter material for broilers, often blended with wood shavings or rice husks[144], and is listed among common bedding materials in the literature[145].

Needs: drying, chopping, and poultry farms nearby. Note that bagasse is normally used *blended* rather than alone — trials ran 50% bagasse with 50% wood shavings or rice husks[145] — and it is listed as one of several common bedding materials rather than a superior one[146].

Read: *A simple, honest sale* — lower value per tonne than fuel briquettes but almost no processing. Best as a use for off-spec or over-moisture material that will not briquette.

4.13 Route 13 — Bioplastics (PHA)

What: ferment bagasse-derived sugars with bacteria to make polyhydroxyalkanoate, a biodegradable plastic.

Evidence: Using residual sugar from bagasse as the substrate for PHA production has been demonstrated directly[59], with the remainder of the lignocellulose available for other processes in a cascading design[59]. Reviews treat bagasse as an attractive PHA feedstock globally[64], and PHA-producing strains have been characterised on sugarcane-derived streams including bagasse[62].

Needs: fermentation vessels, strain handling, extraction, and — the binding constraint — a buyer, because PHA currently sells at a premium and competes with conventional plastic on price, not performance.

Read: *Right destination, wrong starting point.* It is the most elegant scientific route on this list and almost certainly the least bankable for a fibre supplier in 2026.

5 Comparison: where the money and the difficulty actually are

Costs are indicative order-of-magnitude categories, not quotes. Value bands reflect the sources cited in each route's section.

Route	Capital	Technical risk	Time to cash	Read for a fibre supplier
Briquettes / carbonised fuel	Low–Med	Low	Weeks	**Best first move**: proven in Kenya, cash market, solves seasonality
BSF larvae + frass	Med	Med	1–2 months	**Strong**: matches Kenya's protein deficit; needs a second feedstock
Ruminant feed (urea-treated)	Low	Low	Weeks	Good if cattle nearby; maintenance feed, not growth
Anaerobic digestion	Med–High	Med–High	Months	Weak standalone; strong as a mill companion
Biochar (soil / filter)	Med	Low–Med	Weeks–months	Strong second route; shares the kiln with briquettes
2G bioethanol	Very high	High	Years	Not for a supplier; estates are building it
Sugarcane wax / policosanol	High	High	Months–years	Mill-side; press mud is not yours
Board / composites	Med	Med	Months	Plausible locally; blend with wood for quality
Paper / pulp	Very high	High	Years	No, at this scale
Nanocellulose	Very high	Very high	Years	No
Mushrooms	Low–Med	Med	4–8 weeks	Good smallholder cash cycle; it is a food business
Vermicompost	Low	Low	1–3 months	Complementary; pair with BSF
Activated carbon	Med	Med	Months	Depends entirely on a local effluent buyer
Poultry litter	Very low	Very low	Days	Simple sale for off-spec fibre
PHA bioplastic	Very high	Very high	Years	Elegant, not bankable yet

6 The failure modes, stated plainly

This section exists because most waste-to-value projects die from one of four causes, and all four are visible in the sources above.

1. Seasonality. A bagasse-only plant stops when the crush stops. Mumias exported 26 MW to the grid and then *"export to the grid was stopped"*[56] because *"Reliance on bagasse alone as the fuel placed the plant at the mercy of the"* season[56]. HC&S burned 57,100 tons of coal in 2014 to keep its boilers running[100]. Any plan that assumes year-round output from a seasonal feedstock is already broken. Densification, storage, or a second fuel is not optional.

2. Pretreatment cost. Every high-value conversion route — ethanol, PHA, nanocellulose — begins with breaking lignin. The literature is explicit that pretreatment is where the yield changes happen[41][40], and it is also where the capital goes.

3. No buyer. Bagasse biochar, PHA, and nanocellulose are all technically demonstrated and all constrained by market demand rather than process capability[64][76]. A product with no buyer within trucking distance is a very expensive way to store carbon.

4. Feedstock ownership. Wax and press-mud routes belong to whoever runs the clarifiers[71][75]. If your friend supplies fibre rather than owning the mill, several of these routes are not his to take — partnership, not competition, is the move.

7 What I would actually do first

Not a recommendation to spend money — a suggested order of enquiry, chosen to buy information cheaply before committing capital.

1. **Measure the pile.** Tonnes per day, moisture content, and how much is already burned by the mill versus genuinely surplus. The Hawaiian numbers give a template for what a proper inventory looks like[117].

2. **Test briquettes.** Drying plus a small briquette press is the cheapest route to a saleable product, it has an established Kenyan precedent [125][127], and the charcoal market is already liquid. Sell a tonne before building a plant.

3. **Run a BSF pilot on bagasse + a nitrogen stream.** Ten square metres, a few kilos of larvae, and a scale. The literature says bagasse *with* vinasse works best[2], and Kenya's protein gap is documented[89]. Weigh what comes out.

4. **Talk to the mill about cogeneration economics — as a supplier.** Kenya has already proven 26 MW on a PPA[51] and already proven how it fails without a second fuel[56]. Ask what they need and what they would pay per tonne; that conversation is free and it prices every other option.

5. **Only then consider biochar, board, or mushrooms** — each needs a specific buyer to exist first.

One honest caveat about the whole document: price bands for bagasse products vary enormously by market and grade — quoted figures range from cents per kilogram for raw material up to several hundred dollars per metric ton for processed pulp[82][84]. Any business case must be built on a named local buyer and a price they will sign, not on a global range.

Limitations

This is a literature and document survey, not engineering due diligence. No route here has been costed for your friend's specific site, water supply, transport distance, or electricity tariff. Yield figures are from studies conducted in other countries and climates (Brazil, India, Bangladesh, China, Hawaii) and transfer imperfectly; Kenyan-specific numbers are cited as such where they exist. Financial figures are indicative categories rather than quotations, and one market-price range is explicitly flagged as spanning several orders of magnitude[82]. Nothing in this document should be used to make an investment decision without a site-specific feasibility study.

Sources

Every entry below was retrieved during this research. Where a verbatim supporting sentence was captured during retrieval, it is quoted beneath the link; the remainder were cited from their fetched page content.

[**2**] Comparison of Growth and Composition of Black Soldier ...
https://pmc.ncbi.nlm.nih.gov/articles/PMC11508635
"the combination of bagasse and vinasse (BV) proved to be the most suitable substrate among the sugarcane by-products for BSF larva development"

"The UCL treatment also resulted in the larvae's highest protein and oil content."

"Overall, the BSF larvae exhibited the best growth on the UCL substrate, while the combination of bagasse and vinasse (BV) proved to be the most suitable substrate among the sugarcane by-products for BSF larva development."

[**3**] Biowaste and by-products as rearing substrates for black soldier fly (Hermetia illucens) larvae: Effects
https://pmc.ncbi.nlm.nih.gov/articles/PMC9521838

[**4**] Black Soldier Fly larvae: cultivation and its potential as an ...
https://www.lrrd.org/lrrd36/2/3617emma.html
"All feed treatments resulted in BSF larvae with high protein values ranging from 30.6% to 52.2%, with the highest value recorded in the CM-TP treatment."

[**8**] Effect of Wet, Hemi-Solid, and Solid-State Conditions and
https://research-repository.griffith.edu.au/server/api/core/bitstreams/cd169eb3-2a73-4215-9d44-dd6daac0e2bd/content
"Sugarcane bagasse (SCB) constitutes up to 28% of the weight of crushed sugarcane,"

[**9**] Critical Assessment of Hydrogen and Methane Production from 1G and 2G Sugarcane Processing Wastes Using O
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[**12**] [PDF] An Overview of Biogas Production from Anaerobic Digestion and the ...
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[**14**] Sugarcane bagasse biochar increases soil carbon sequestration and yields of maize and groundnut in charla
https://www.tandfonline.com/doi/full/10.1080/03650340.2021.1892651

[**18**] Sugarcane bagasse-based biochar and its potential applications: a review
https://link.springer.com/article/10.1007/s42247-023-00603-y
"Recognizing the importance of addressing this issue, researchers have identified the conversion of agricultural waste into biochar as an efficient means of harnessing energy following biomass devolatilization."

"Sugarcane bagasse-based biochar and its potential applications: a review"

[**19**] The Effects of Different Substrates on the Growth, Yield, and ...
https://pmc.ncbi.nlm.nih.gov/articles/PMC4731647
"It is also found that the C/N ratio of substrate formulas has close correlation with total colonization period, mushroom weight, yield, BE and protein content of mushroom PO and PC."

[**20**] The Effect of Different Lignocellulose Biomass-based ...
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[**21**] Sugarcane Bagasse-An Ideal Substrate for Oyster Mushroom ...

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"The highest yield (120.00g) and return (Tk 12.00) were obtained from the treatment of 80% of sugarcane bagasse supplemented with 20% wheat bran."

[**22**] © 2019 JETIR June 2019, Volume 6, Issue 6 www.

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[**70**] Policosanol: Extraction and biological functions
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[**71**] Applications of Sugarcane Wax and it’s Products: A Review
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[Price-Watch](https://www.price-watch.ai/bagasse-pulp-prices/)."

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"Bagasse is the fibre left behind after the extraction of juice from sugarcane; it is about 30% of the weight of cane."

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https://mauicauses.org/demise-of-power-purchase-deal-is-final-blow-for-hcs
"HC&S generates electricity by burning the fiber known as bagasse left over from processed cane"

"said its main reasons for ending 146 years of sugar cane farming and shuttering the plantation with 675 employees was two bad years of low sugar production and prices"

"In 2014 the company burned 57,100 tons of coal"

[**104**] HC&S closure will pull plug on power deal
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"Hawaiian Commercial & Sugar Co. provided 1,265.5 megawatt hours of power to Maui Electric Co. in the third quarter"

"It scaled back HC&S power deliveries to MECO significantly from as much as 12 MW of power year-round."

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"Alone among potential ethanol producers in Hawai\{}‘i, Kennett, president of Gay & Robinson, Inc., the Kaua\{}‘i sugar planter, is pursuing a two-pronged ethanol production strategy."

[**117**] [PDF] The Potential for Biofuels Production in Hawai’i - Hawaii Energy Office
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"In 2006, HC&S produced about 430,000 tons of bagasse at 50 percent moisture content (215,000 tons of fiber) and 60,000 tons of molasses."

"The waste streams evaluated in this effort for their biofuel production potential include municipal solid waste, sewage sludge, FOG, bagasse fiber, cane trash, molasses, pineapple processing waste, macadamia nut shells, animal manures, and landfill g"

"The sugar industry is the largest single source of agricultural biomass residues in ## the State of Hawai’i."

[**124**] An innovative approach of developing agro-industrial waste to biofuel value chain to avoid charcoal drive
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[**146**] Effects of using different litter material on performance and ...
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[**149**] A Study on Yield and Gut Flora Diversity on Vermicomposting ...
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[**151**] [PDF] Vermicomposting of sugarcane bagasse using Eudrillus eugeniae
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"This study investigated a simple and eco-friendly way to manage different types of biowaste by using earthworms and biochar (a charcoal-like material made from sugarcane bagasse)."

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